JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Hyperbola

JEE Main Mathematics · Previous Year Questions with Answer Key

97	90	2003 – 2026	30	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 51 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x = \pm\frac{4\sqrt{6}}{3}$. Let $\mathrm{H} : \frac{x^2}{\mathrm{a}^2} - \frac{y^2}{\mathrm{b}^2} = 1$ be a hyperbola whose eccentricity is equal to the length of semi-major axis of E , and whose length of latus rectum is equal to the length of minor axis of E . Then the distance between the foci of H is :

(A) $\frac{4\sqrt{2}}{\sqrt{7}}$ (B) $\frac{4\sqrt{2}}{7}$ (C) $\frac{4}{\sqrt{7}}$ (D) $\frac{8}{7}$

Q2. JEE Main 2026 (Online) 6th April Morning Shift

Let e_1 and e_2 be two distinct roots of the equation $x^2 - ax + 2 = 0$. Let the sets $\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of hyperbolas $\} = (\alpha, \beta)$, and $\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of an ellipse and a hyperbola, respectively $\} = (\gamma, \infty)$.

Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

(A) 18 (B) 22 (C) 26 (D) 34

Q3. JEE Main 2026 (Online) 6th April Morning Shift

If the eccentricity e of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, passing through $(6, 4\sqrt{3})$, satisfies $15\left(e^2 + 1\right) = 34e$, then the length of the latus rectum of the hyperbola $\frac{x^2}{b^2} - \frac{y^2}{2(a^2+1)} = 1$ is:

(A) 10 (B) 20 (C) 25 (D) 30

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Let the eccentricity e of a hyperbola satisfy the equation $6\mathrm{e}^2 - 11\mathrm{e} + 3 = 0$. If the foci of the hyperbola are $(3, 5)$ and $(3, -4)$, then the length of its latus rectum is :

(A) $11/3$ (B) $17/3$ (C) $15/2$ (D) $17/2$

Q5. JEE Main 2026 (Online) 4th April Evening Shift

Let $\mathrm{H} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be a hyperbola such that the distance between its foci is 6 and the distance between its directrices is $\frac{8}{3}$. If the line $x = \alpha$ intersects the hyperbola H at the points A and B such that the area of the triangle AOB is $4\sqrt{15}$, where O is the origin, then α^2 equals

(A) 12 (B) 16 (C) 24 (D) 25

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Let O be the origin, and P and Q be two points on the rectangular hyperbola $xy = 12$ such that the midpoint of the line segment PQ is $\left(\frac{1}{2}, -\frac{1}{2}\right)$. Then the area of the triangle OPQ equals :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

Q7. JEE Main 2026 (Online) 28th January Evening Shift

Let the ellipse $E: \frac{x^2}{144}+\frac{y^2}{169}=1$ and the hyperbola $H: \frac{x^2}{16}-\frac{y^2}{\lambda^2}=-1$ have the same foci. If e and L respectively denote the eccentricity and the length of the latus rectum of H, then the value of $24(e+L)$ is :

(A) 296 (B) 126 (C) 67 (D) 148

Q8. JEE Main 2026 (Online) 23rd January Evening Shift

Let PQ be a chord of the hyperbola $\frac{x^2}{4}-\frac{y^2}{b^2}=1$, perpendicular to the x -axis such that OPQ is an equilateral triangle, O being the centre of the hyperbola. If the eccentricity of the hyperbola is $\sqrt{3}$, then the area of the triangle OPQ is

(A) $2\sqrt{3}$ (B) $\frac{11}{5}$ (C) $\frac{8\sqrt{3}}{5}$ (D) $\frac{9}{5}$

Q9. JEE Main 2026 (Online) 23rd January Morning Shift

Let the domain of the function $f(x)=\log_3\log_5\log_7\left(9x-x^2-13\right)$ be the interval (m,n). Let the hyperbola $\frac{x^2}{\mathrm{a}^2}-\frac{y^2}{\mathrm{b}^2}=1$ have eccentricity $\frac{\mathrm{n}}{3}$ and the length of the latus rectum $\frac{8\,\mathrm{m}}{3}$. Then $\mathrm{b}^2-\mathrm{a}^2$ is equal to :

(A) 7 (B) 9 (C) 11 (D) 5

Q10. JEE Main 2026 (Online) 22nd January Evening Shift

Let $\mathrm{P}(10,2\sqrt{15})$ be a point on the hyperbola $\frac{x^2}{\mathrm{a}^2}-\frac{y^2}{\mathrm{b}^2}=1$, whose foci are S and S'. If the length of its latus rectum is 8 , then the square of the area of $\Delta\mathrm{PSS}'$ is equal to :

(A) 4200 (B) 1462 (C) 900 (D) 2700

Q11. JEE Main 2026 (Online) 22nd January Morning Shift

If the line $\alpha x+2y=1$, where $\alpha\in\mathbb{R}$, does not meet the hyperbola $x^2-9y^2=9$, then a possible value of α is :

(A) 0.6 (B) 0.7 (C) 0.8 (D) 0.5

Q12. JEE Main 2026 (Online) 21st January Morning Shift

Let the foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$. If the eccentricity of the hyperbola is 5 , then the length of its latus rectum is :

(A) $\frac{96}{\sqrt{5}}$ (B) $24\sqrt{5}$ (C) 12 (D) 16

Q13. JEE Main 2025 (Online) 7th April Evening Shift

Let e_1 and e_2 be the eccentricities of the ellipse $\frac{x^2}{b^2}+\frac{y^2}{25}=1$ and the hyperbola $\frac{x^2}{16}-\frac{y^2}{b^2}=1$, respectively. If $b < 5$ and $e_1e_2 = 1$, then the eccentricity of the ellipse having its axes along the coordinate axes and passing through all four foci (two of the ellipse and two of the hyperbola) is :

(A) $\frac{4}{5}$ (B) $\frac{3}{5}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{3}}{2}$

Q14. JEE Main 2025 (Online) 4th April Evening Shift

Let the sum of the focal distances of the point $\mathrm{P}(4,3)$ on the hyperbola $\mathrm{H}: \frac{x^2}{\mathrm{a}^2}-\frac{y^2}{\mathrm{b}^2}=1$ be $8\sqrt{\frac{5}{3}}$. If for H , the length of the latus rectum is l and the product of the focal distances of the point P is m , then $9l^2+6\,\mathrm{m}$ is equal to :

(A) 187 (B) 184 (C) 186 (D) 185

Q15. JEE Main 2025 (Online) 2nd April Morning Shift

Let one focus of the hyperbola $\mathrm{H}: \frac{x^2}{\mathrm{a}^2}-\frac{y^2}{\mathrm{b}^2}=1$ be at $(\sqrt{10},0)$ and the corresponding directrix be $x=\frac{9}{\sqrt{10}}$. If e and l respectively are the eccentricity and the length of the latus rectum of H , then $9\left(e^2+l\right)$ is equal to :

(A) 12 (B) 14 (C) 15 (D) 16

Q16. JEE Main 2025 (Online) 22nd January Morning Shift

Let the foci of a hyperbola be $(1,14)$ and $(1,-12)$. If it passes through the point $(1,6)$, then the length of its latus-rectum is :

(A) $\frac{25}{6}$ (B) $\frac{144}{5}$ (C) $\frac{288}{5}$ (D) $\frac{24}{5}$

Q17. JEE Main 2024 (Online) 9th April Evening Shift

Let the foci of a hyperbola H coincide with the foci of the ellipse $E: \frac{(x-1)^2}{100}+\frac{(y-1)^2}{75}=1$ and the eccentricity of the hyperbola H be the reciprocal of the eccentricity of the ellipse E. If the length of the transverse axis of H is α and the length of its conjugate axis is β, then $3\alpha^2+2\beta^2$ is equal to

(A) 225 (B) 237 (C) 242 (D) 205

Q18. JEE Main 2024 (Online) 8th April Morning Shift

Let $H: \frac{-x^2}{a^2}+\frac{y^2}{b^2}=1$ be the hyperbola, whose eccentricity is $\sqrt{3}$ and the length of the latus rectum is $4\sqrt{3}$. Suppose the point $(\alpha,6), \alpha>0$ lies on H. If β is the product of the focal distances of the point $(\alpha,6)$, then $\alpha^2+\beta$ is equal to

(A) 170 (B) 171 (C) 169 (D) 172

Q19. JEE Main 2024 (Online) 4th April Evening Shift

Consider a hyperbola H having centre at the origin and foci on the x-axis. Let C_1 be the circle touching the hyperbola H and having the centre at the origin. Let C_2 be the circle touching the hyperbola H at its vertex and having the centre at one of its foci. If areas (in sq units) of C_1 and C_2 are 36π and 4π, respectively, then the length (in units) of latus rectum of H is

(A) $\frac{28}{3}$ (B) $\frac{11}{3}$ (C) $\frac{14}{3}$ (D) $\frac{10}{3}$

Q20. JEE Main 2024 (Online) 1st February Morning Shift

For $0 < \theta < \pi/2$, if the eccentricity of the hyperbola

$x^2 - y^2 \operatorname{cosec}^2\theta = 5$ is $\sqrt{7}$ times eccentricity of the

ellipse $x^2 \operatorname{cosec}^2\theta + y^2 = 5$, then the value of θ is :

(A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{12}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Q21. JEE Main 2024 (Online) 31st January Morning Shift

If the foci of a hyperbola are same as that of the ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$ and the eccentricity of the hyperbola is $\frac{15}{8}$ times the eccentricity of the ellipse, then the smaller focal distance of the point $\left(\sqrt{2}, \frac{14}{3}\sqrt{\frac{2}{5}}\right)$ on the hyperbola, is equal to

(A) $14\sqrt{\frac{2}{5}} - \frac{4}{3}$ (B) $7\sqrt{\frac{2}{5}} + \frac{8}{3}$

(C) $7\sqrt{\frac{2}{5}} - \frac{8}{3}$ (D) $14\sqrt{\frac{2}{5}} - \frac{16}{3}$

Q22. JEE Main 2024 (Online) 30th January Evening Shift

Let P be a point on the hyperbola $H : \frac{x^2}{9} - \frac{y^2}{4} = 1$, in the first quadrant such that the area of triangle formed by P and the two foci of H is $2\sqrt{13}$. Then, the square of the distance of P from the origin is

(A) 26 (B) 22 (C) 20 (D) 18

Q23. JEE Main 2024 (Online) 27th January Evening Shift

Let e_1 be the eccentricity of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and e_2 be the eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \mathrm{a} > \mathrm{b}$, which passes through the foci of the hyperbola. If $\mathrm{e_1 e_2} = 1$, then the length of the chord of the ellipse parallel to the x-axis and passing through $(0, 2)$ is :

(A) $\frac{8\sqrt{5}}{3}$ (B) $3\sqrt{5}$ (C) $4\sqrt{5}$ (D) $\frac{10\sqrt{5}}{3}$

Q24. JEE Main 2023 (Online) 11th April Morning Shift

Let R be a rectangle given by the lines $x = 0, x = 2, y = 0$ and $y = 5$. Let A$(\alpha, 0)$ and B$(0, \beta), \alpha \in [0, 2]$ and $\beta \in [0, 5]$, be such that the line segment AB divides the area of the rectangle R in the ratio 4 : 1. Then, the mid-point of AB lies on a :

(A) hyperbola (B) straight line (C) parabola (D) circle

Q25. JEE Main 2023 (Online) 31st January Evening Shift

Let H be the hyperbola, whose foci are $(1 \pm \sqrt{2}, 0)$ and eccentricity is $\sqrt{2}$. Then the length of its latus rectum is :

(A) $\frac{5}{2}$ (B) 3 (C) 2 (D) $\frac{3}{2}$

Q26. JEE Main 2022 (Online) 28th July Evening Shift

Let the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ pass through the point $(2\sqrt{2}, -2\sqrt{2})$. A parabola is drawn whose focus is same as the focus of H with positive abscissa and the directrix of the parabola passes through the other focus of H. If the length of the latus rectum of the parabola is e times the length of the latus rectum of H, where e is the eccentricity of H, then which of the following points lies on the parabola?

(A) $(2\sqrt{3}, 3\sqrt{2})$ (B) $(3\sqrt{3}, -6\sqrt{2})$ (C) $(\sqrt{3}, -\sqrt{6})$ (D) $(3\sqrt{6}, 6\sqrt{2})$

Q27. JEE Main 2022 (Online) 26th July Evening Shift

If the line $x - 1 = 0$ is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point :

(A) $(-2\sqrt{5}, 6)$ (B) $(-\sqrt{5}, 3)$ (C) $(\sqrt{5}, -2)$ (D) $(2\sqrt{5}, 3\sqrt{6})$

Q28. JEE Main 2022 (Online) 25th July Evening Shift

Let the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{7} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{\alpha} = \frac{1}{25}$ coincide. Then the length of the latus rectum of the hyperbola is :

(A) $\frac{32}{9}$ (B) $\frac{18}{5}$ (C) $\frac{27}{4}$ (D) $\frac{27}{10}$

Q29. JEE Main 2022 (Online) 28th June Evening Shift

Let a > 0, b > 0. Let e and l respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Let e' and l' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2 = \frac{11}{14}l$ and $(e')^2 = \frac{11}{8}l'$, then the value of $77a + 44b$ is equal to :

(A) 100 (B) 110 (C) 120 (D) 130

Q30. JEE Main 2021 (Online) 26th August Evening Shift

The locus of the mid points of the chords of the hyperbola $x^2 - y^2 = 4$, which touch the parabola $y^2 = 8x$, is :

(A) $y^3(x - 2) = x^2$ (B) $x^3(x - 2) = y^2$ (C) $y^2(x - 2) = x^3$ (D) $x^2(x - 2) = y^3$

Q31. JEE Main 2021 (Online) 25th July Morning Shift

The locus of the centroid of the triangle formed by any point P on the hyperbola $16x^2 - 9y^2 + 32x + 36y - 164 = 0$, and its foci is :

(A) $16x^2 - 9y^2 + 32x + 36y - 36 = 0$

(B) $9x^2 - 16y^2 + 36x + 32y - 144 = 0$

(C) $16x^2 - 9y^2 + 32x + 36y - 144 = 0$

(D) $9x^2 - 16y^2 + 36x + 32y - 36 = 0$

Q32. JEE Main 2021 (Online) 16th March Morning Shift

The locus of the midpoints of the chord of the circle, $x^2 + y^2 = 25$ which is tangent to the hyperbola, $\frac{x^2}{9} - \frac{y^2}{16} = 1$ is :

(A) $(x^2 + y^2)^2 - 9x^2 + 16y^2 = 0$

(B) $(x^2 + y^2)^2 - 9x^2 + 144y^2 = 0$

(C) $(x^2 + y^2)^2 - 16x^2 + 9y^2 = 0$

(D) $(x^2 + y^2)^2 - 9x^2 - 16y^2 = 0$

Q33. JEE Main 2021 (Online) 25th February Evening Shift

A hyperbola passes through the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and its transverse and conjugate axes coincide with major and minor axes of the ellipse, respectively. If the product of their eccentricities is one, then the equation of the hyperbola is :

(A) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

(B) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(C) $\frac{x^2}{9} - \frac{y^2}{25} = 1$

(D) $x^2 - y^2 = 9$

Q34. JEE Main 2020 (Online) 3rd September Evening Slot

Let e_1 and e_2 be the eccentricities of the ellipse, $\frac{x^2}{25} + \frac{y^2}{b^2} = 1 (b < 5)$ and the hyperbola, $\frac{x^2}{16} - \frac{y^2}{b^2} = 1$ respectively satisfying $e_1e_2 = 1$. If α and β are the distances between the foci of the ellipse and the foci of the hyperbola respectively, then the ordered pair (α, β) is equal to :

(A) (8, 10)

(B) (8, 12)

(C) $\left(\frac{24}{5}, 10\right)$

(D) $\left(\frac{20}{3}, 12\right)$

Q35. JEE Main 2020 (Online) 3rd September Morning Slot

A hyperbola having the transverse axis of length $\sqrt{2}$ has the same foci as that of the ellipse $3x^2 + 4y^2 = 12$, then this hyperbola does not pass through which of the following points?

(A) $\left(1, -\frac{1}{\sqrt{2}}\right)$

(B) $\left(\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}\right)$

(C) $\left(-\sqrt{\frac{3}{2}}, 1\right)$

(D) $\left(\frac{1}{\sqrt{2}}, 0\right)$

Q36. JEE Main 2020 (Online) 2nd September Evening Slot

For some $\theta \in \left(0, \frac{\pi}{2}\right)$, if the eccentricity of the hyperbola, $x^2 - y^2\sec^2\theta = 10$ is $\sqrt{5}$ times the eccentricity of the ellipse, $x^2\sec^2\theta + y^2 = 5$, then the length of the latus rectum of the ellipse, is :

(A) $\sqrt{30}$ (B) $2\sqrt{6}$ (C) $\frac{4\sqrt{5}}{3}$ (D) $\frac{2\sqrt{5}}{3}$

Q37. JEE Main 2020 (Online) 9th January Morning Slot

If e_1 and e_2 are the eccentricities of the ellipse, $\frac{x^2}{18} + \frac{y^2}{4} = 1$ and the hyperbola, $\frac{x^2}{9} - \frac{y^2}{4} = 1$ respectively and (e_1, e_2) is a point on the ellipse, $15x^2 + 3y^2 = k$, then k is equal to :

(A) 17 (B) 16 (C) 15 (D) 14

Q38. JEE Main 2019 (Online) 10th April Evening Slot

If $5x + 9 = 0$ is the directrix of the hyperbola $16x^2 - 9y^2 = 144$, then its corresponding focus is :

(A) $\left(\frac{5}{3}, 0\right)$ (B) (5, 0)

(C) (- 5, 0) (D) $\left(-\frac{5}{3}, 0\right)$

Q39. JEE Main 2019 (Online) 10th April Morning Slot

If a directrix of a hyperbola centred at the origin and passing through the point $(4, -2\sqrt{3})$ is $5x = 4\sqrt{5}$ and its eccentricity is e, then :

(A) $4e^4 - 24e^2 + 27 = 0$ (B) $4e^4 - 24e^2 + 35 = 0$

(C) $4e^4 - 12e^2 - 27 = 0$ (D) $4e^4 + 8e^2 - 35 = 0$

Q40. JEE Main 2019 (Online) 12th January Morning Slot

If the vertices of a hyperbola be at (−2, 0) and (2, 0) and one of its foci be at (−3, 0), then which one of the following points does not lie on this hyperbola?

(A) $\left(6, 5\sqrt{2}\right)$ (B) $\left(2\sqrt{6}, 5\right)$

(C) $\left(-6, 2\sqrt{10}\right)$ (D) $\left(4, \sqrt{15}\right)$

Q41. JEE Main 2019 (Online) 11th January Evening Slot

If a hyperbola has length of its conjugate axis equal to 5 and the distance between its foci is 13, then the eccentricity of the hyperbola is :

(A) $\frac{13}{6}$ (B) 2 (C) $\frac{13}{12}$ (D) $\frac{13}{8}$

Q42. JEE Main 2019 (Online) 9th January Evening Slot

A hyperbola has its centre at the origin, passes through the point (4, 2) and has transverse axis of length 4 along the x-axis. Then the eccentricity of the hyperbola is :

(A) $\frac{3}{2}$ (B) $\sqrt{3}$ (C) 2 (D) $\frac{2}{\sqrt{3}}$

Q43. JEE Main 2019 (Online) 9th January Morning Slot

Let $0 < \theta < \frac{\pi}{2}$. If the eccentricity of the

hyperbola $\frac{x^2}{\cos^2\theta} - \frac{y^2}{\sin^2\theta}$ = 1 is greater

than 2, then the length of its latus rectum lies in the interval :

(A) $(3, \infty)$ (B) $\left(\frac{3}{2}, 2\right]$ (C) $\left(1, \frac{3}{2}\right]$ (D) $(2, 3]$

Q44. JEE Main 2018 (Online) 16th April Morning Slot

The locus of the point of intersection of the lines, $\sqrt{2}x - y + 4\sqrt{2}k = 0$ and $\sqrt{2}kx + ky - 4\sqrt{2} = 0$ (k is any non-zero real parameter), is :

(A) an ellipse whose eccentricity is $\frac{1}{\sqrt{3}}$.

(B) an ellipse with length of its major axis $8\sqrt{2}$.

(C) a hyperbola whose eccentricity is $\sqrt{3}$.

(D) a hyperbola with length of its transverse axis $8\sqrt{2}$.

Q45. JEE Main 2018 (Online) 15th April Morning Slot

If the tangents drawn to the hyperbola $4y^2 = x^2 + 1$ intersect the co-ordinate axes at the distinct points A and B then the locus of the mid point of AB is :

(A) $x^2 - 4y^2 + 16x^2y^2 = 0$ (B) $x^2 - 4y^2 - 16x^2y^2 = 0$

(C) $4x^2 - y^2 + 16x^2y^2 = 0$ (D) $4x^2 - y^2 - 16x^2y^2 = 0$

Q46. JEE Main 2017 (Online) 8th April Morning Slot

The locus of the point of intersection of the straight lines,

$tx - 2y - 3t = 0$

$x - 2ty + 3 = 0$ **($t \in R$)**, is :

(A) an ellipse with eccentricity $\frac{2}{\sqrt{5}}$

(B) an ellipse with the length of major axis 6

(C) a hyperbola with eccentricity $\sqrt{5}$

(D) a hyperbola with the length of conjugate axis 3

Q47. JEE Main 2016 (Online) 10th April Morning Slot

A hyperbola whose transverse axis is along the major axis of the conic, $\frac{x^2}{3}+\frac{y^2}{4}=4$ and has vertices at the foci of this conic. If the eccentricity of the hyperbola is $\frac{3}{2}$, then which of the following points does **NOT** lie on it?

(A) $(0, 2)$ (B) $\left(\sqrt{5}, 2\sqrt{2}\right)$

(C) $\left(\sqrt{10}, 2\sqrt{3}\right)$ (D) $\left(5, 2\sqrt{3}\right)$

Q48. JEE Main 2016 (Online) 9th April Morning Slot

Let a and b respectively be the semitransverse and semi-conjugate axes of a hyperbola whose eccentricity satisfies the equation $9e^2 - 18e + 5 = 0$. If S(5, 0) is a focus and $5x = 9$ is the corresponding directrix of this hyperbola, then $a^2 - b^2$ is equal to :

(A) 7 (B) -7 (C) 5 (D) -5

Q49. JEE Main 2016 (Offline)

The eccentricity of the hyperbola whose length of the latus rectum is equal to 8 and the length of its conjugate axis is equal to half of the distance between its foci, is :

(A) $\frac{2}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) $\frac{4}{3}$ (D) $\frac{4}{\sqrt{3}}$

Q50. AIEEE 2007

For the Hyperbola $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$, which of the following remains constant when α varies=?

(A) abscissae of vertices (B) abscissae of foci

(C) eccentricity (D) directrix.

Q51. AIEEE 2003

The foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide. Then the value of b^2 is :

(A) 9 (B) 1 (C) 5 (D) 7

Numerical 16 questions

Q52. JEE Main 2026 (Online) 28th January Morning Shift

For some $\theta \in \left(0, \frac{\pi}{2}\right)$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2 - y^2\sec^2\theta = 8$ be e_1 and l_1, respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2\sec^2\theta + y^2 = 6$ be e_2 and l_2, respectively. If $e_1^2 = e_2^2\left(\sec^2\theta + 1\right)$, then $\left(\frac{l_1 l_2}{e_1 e_2}\right)\tan^2\theta$ is equal to

Q53. JEE Main 2025 (Online) 7th April Evening Shift

Let the lengths of the transverse and conjugate axes of a hyperbola in standard form be $2a$ and $2b$, respectively, and one focus and the corresponding directrix of this hyperbola be $(-5, 0)$ and $5x + 9 = 0$, respectively. If the product of the focal distances of a point $(\alpha, 2\sqrt{5})$ on the hyperbola is p, then $4p$ is equal to _____.

Q54. JEE Main 2025 (Online) 7th April Morning Shift

Consider the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ having one of its focus at $\mathrm{P}(-3, 0)$. If the latus rectum through its other focus subtends a right angle at P and $a^2b^2 = \alpha\sqrt{2} - \beta, \alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is _____ .

Q55. JEE Main 2025 (Online) 3rd April Evening Shift

If the equation of the hyperbola with foci $(4, 2)$ and $(8, 2)$ is $3x^2 - y^2 - \alpha x + \beta y + \gamma = 0$, then $\alpha + \beta + \gamma$ is equal to_____.

Q56. JEE Main 2025 (Online) 3rd April Morning Shift

Let the product of the focal distances of the point $\mathbf{P}(4, 2\sqrt{3})$ on the hyperbola $\mathrm{H} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be 32 . Let the length of the conjugate axis of H be p and the length of its latus rectum be q. Then $p^2 + q^2$ is equal to_____

Q57. JEE Main 2025 (Online) 24th January Evening Shift

Let $\mathrm{H}_1 : \frac{x^2}{\mathrm{a}^2} - \frac{y^2}{\mathrm{b}^2} = 1$ and $\mathrm{H}_2 : -\frac{x^2}{\mathrm{A}^2} + \frac{y^2}{\mathrm{B}^2} = 1$ be two hyperbolas having length of latus rectums $15\sqrt{2}$ and $12\sqrt{5}$ respectively. Let their eccentricities be $e_1 = \sqrt{\frac{5}{2}}$ and e_2 respectively. If the product of the lengths of their transverse axes is $100\sqrt{10}$, then $25\mathrm{e}_2^2$ is equal to _____ .

Q58. JEE Main 2024 (Online) 8th April Evening Shift

Let S be the focus of the hyperbola $\frac{x^2}{3} - \frac{y^2}{5} = 1$, on the positive x-axis. Let C be the circle with its centre at $\mathrm{A}(\sqrt{6}, \sqrt{5})$ and passing through the point S. If O is the origin and SAB is a diameter of C, then the square of the area of the triangle OSB is equal to _____.

Q59. JEE Main 2024 (Online) 6th April Evening Shift

The length of the latus rectum and directrices of hyperbola with eccentricity e are 9 and $x = \pm\frac{4}{\sqrt{13}}$, respectively. Let the line $y - \sqrt{3}x + \sqrt{3} = 0$ touch this hyperbola at (x_0, y_0). If m is the product of the focal distances of the point (x_0, y_0), then $4\mathrm{e}^2 + \mathrm{m}$ is equal to _____.

Q60. JEE Main 2024 (Online) 31st January Morning Shift

Let the foci and length of the latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ be $(\pm 5, 0)$ and $\sqrt{50}$, respectively. Then, the square of the eccentricity of the hyperbola $\frac{x^2}{b^2} - \frac{y^2}{a^2b^2} = 1$ equals

Q61. JEE Main 2024 (Online) 30th January Morning Shift

Let the latus rectum of the hyperbola $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$ subtend an angle of $\frac{\pi}{3}$ at the centre of the hyperbola. If b^2 is equal to $\frac{l}{\mathrm{m}}(1 + \sqrt{\mathrm{n}})$, where l and m are co-prime numbers, then $\mathrm{l}^2 + \mathrm{m}^2 + \mathrm{n}^2$ is equal to _____.

Q62. JEE Main 2023 (Online) 11th April Morning Shift

Let $\mathrm{H_n} : \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1, n \in N$. Let k be the smallest even value of n such that the eccentricity of $\mathrm{H_k}$ is a rational number. If l is the length of the latus rectum of $\mathrm{H_k}$, then $21l$ is equal to _____.

Q63. JEE Main 2023 (Online) 6th April Evening Shift

Let the eccentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is reciprocal to that of the hyperbola $2x^2 - 2y^2 = 1$. If the ellipse intersects the hyperbola at right angles, then square of length of the latus-rectum of the ellipse is _____.

Q64. JEE Main 2022 (Online) 27th July Morning Shift

An ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the vertices of the hyperbola $H : \frac{x^2}{49} - \frac{y^2}{64} = -1$. Let the major and minor axes of the ellipse E coincide with the transverse and conjugate axes of the hyperbola H, respectively. Let the product of the eccentricities of E and H be $\frac{1}{2}$. If l is the length of the latus rectum of the ellipse E, then the value of $113l$ is equal to _____.

Q65. JEE Main 2022 (Online) 29th June Morning Shift

Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, a > 0, b > 0, be a hyperbola such that the sum of lengths of the transverse and the conjugate axes is $4(2\sqrt{2} + \sqrt{14})$. If the eccentricity H is $\frac{\sqrt{11}}{2}$, then the value of $a^2 + b^2$ is equal to _____.

Q66. JEE Main 2022 (Online) 24th June Evening Shift

Let the hyperbola $H : \frac{x^2}{a^2} - y^2 = 1$ and the ellipse $E : 3x^2 + 4y^2 = 12$ be such that the length of latus rectum of H is equal to the length of latus rectum of E. If e_H and e_E are the eccentricities of H and E respectively, then the value of $12\left(e_H^2 + e_E^2\right)$ is equal to _____.

Q67. JEE Main 2021 (Online) 25th February Morning Shift

The locus of the point of intersection of the lines $\left(\sqrt{3}\right)kx + ky - 4\sqrt{3} = 0$ and $\sqrt{3}x - y - 4\left(\sqrt{3}\right)k = 0$ is a conic, whose eccentricity is _____.

Out of syllabus 30 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q68. JEE Main 2023 (Online) 1st February Evening Shift · MCQ (Single Correct Answer)

Let $\mathrm{P}(x_0, y_0)$ be the point on the hyperbola $3x^2 - 4y^2 = 36$, which is nearest to the line $3x + 2y = 1$. Then $\sqrt{2}(y_0 - x_0)$ is equal to :

(A) 3 (B) -9 (C) -3 (D) 9

Q69. JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

Let the tangent drawn to the parabola $y^2 = 24x$ at the point (α, β) is perpendicular to the line $2x + 2y = 5$. Then the normal to the hyperbola $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ at the point $(\alpha + 4, \beta + 4)$ does NOT pass through the point :

(A) (25, 10) (B) (20, 12) (C) (30, 8) (D) (15, 13)

Q70. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

Let the eccentricity of the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $\sqrt{\frac{5}{2}}$ and length of its latus rectum be $6\sqrt{2}$. If $y = 2x + c$ is a tangent to the hyperbola H, then the value of c^2 is equal to :

(A) 18 (B) 20 (C) 24 (D) 32

Q71. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

The normal to the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{9} = 1$ at the point $\left(8, 3\sqrt{3}\right)$ on it passes through the point :

(A) $\left(15, -2\sqrt{3}\right)$ (B) $\left(9, 2\sqrt{3}\right)$ (C) $\left(-1, 9\sqrt{3}\right)$ (D) $\left(-1, 6\sqrt{3}\right)$

Q72. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

Let $\lambda x - 2y = \mu$ be a tangent to the hyperbola $a^2x^2 - y^2 = b^2$. Then $\left(\frac{\lambda}{a}\right)^2 - \left(\frac{\mu}{b}\right)^2$ is equal to :

(A) -2 (B) -4 (C) 2 (D) 4

Q73. JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

The point $P\left(-2\sqrt{6}, \sqrt{3}\right)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ having eccentricity $\frac{\sqrt{5}}{2}$. If the tangent and normal at P to the hyperbola intersect its conjugate axis at the point Q and R respectively, then QR is equal to :

(A) $4\sqrt{3}$ (B) 6 (C) $6\sqrt{3}$ (D) $3\sqrt{6}$

Q74. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Let a line L : 2x + y = k, k > 0 be a tangent to the hyperbola $x^2 - y^2 = 3$. If L is also a tangent to the parabola $y^2 = \alpha x$, then α is equal to :

(A) 12 (B) -12 (C) 24 (D) -24

Q75. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

Consider a hyperbola H : $x^2 - 2y^2 = 4$. Let the tangent at a point P(4, $\sqrt{6}$) meet the x-axis at Q and latus rectum at R(x_1, y_1), $x_1 > 0$. If F is a focus of H which is nearer to the point P, then the area of ΔQFR is equal to :

(A) $\sqrt{6} - 1$ (B) $\frac{7}{\sqrt{6}} - 2$ (C) $4\sqrt{6} - 1$ (D) $4\sqrt{6}$

Q76. JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

If the line y = mx + c is a common tangent to the hyperbola $\frac{x^2}{100} - \frac{y^2}{64} = 1$ and the circle $x^2 + y^2 = 36$, then which one of the following is true?

(A) 5m = 4 (B) 8m + 5 = 0 (C) $c^2 = 369$ (D) $4c^2 = 369$

Q77. JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

Let P(3, 3) be a point on the hyperbola, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal to it at P intersects the x-axis at (9, 0) and e is its eccentricity, then the ordered pair (a^2, e^2) is equal to :

(A) $\left(\frac{9}{2}, 2\right)$ (B) $\left(\frac{3}{2}, 2\right)$ (C) (9,3) (D) $\left(\frac{9}{2}, 3\right)$

Q78. JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

A line parallel to the straight line 2x − y = 0 is tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ at the point (x_1, y_1). Then $x_1^2 + 5y_1^2$ is equal to :

(A) 5 (B) 6 (C) 10 (D) 8

Q79. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

If a hyperbola passes through the point P(10, 16) and it has vertices at (± 6, 0), then the equation of the normal to it at P is :

(A) 2x + 5y = 100 (B) x + 3y = 58 (C) x + 2y = 42 (D) 3x + 4y = 94

Q80. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the line y = mx + $7\sqrt{3}$ is normal to the hyperbola $\frac{x^2}{24} - \frac{y^2}{18} = 1$, then a value of m is :

(A) $\frac{3}{\sqrt{5}}$ (B) $\frac{\sqrt{15}}{2}$ (C) $\frac{\sqrt{5}}{2}$ (D) $\frac{2}{\sqrt{5}}$

Q81. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

If the eccentricity of the standard hyperbola passing through the point (4,6) is 2, then the equation of the tangent to the hyperbola at (4,6) is :

(A) 2x − y − 2 = 0 (B) 3x − 2y = 0 (C) 2x − 3y + 10 = 0 (D) x − 2y + 8 = 0

Q82. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

Equation of a common tangent to the parabola $y^2 = 4x$ and the hyperbola $xy = 2$ is :

(A) $x + y + 1 = 0$ (B) $4x + 2y + 1 = 0$ (C) $x - 2y + 4 = 0$ (D) $x + 2y + 4 = 0$

Q83. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

The equation of a tangent to the hyperbola $4x^2 - 5y^2 = 20$ parallel to the line $x - y = 2$ is :

(A) $x - y + 9 = 0$ (B) $x - y - 3 = 0$ (C) $x - y + 1 = 0$ (D) $x - y + 7 = 0$

Q84. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

Tangents are drawn to the hyperbola $4x^2 - y^2 = 36$ at the points P and Q.

If these tangents intersect at the point T(0, 3) then the area (in sq. units) of ΔPTQ is :

(A) $36\sqrt{5}$ (B) $45\sqrt{5}$ (C) $54\sqrt{3}$ (D) $60\sqrt{3}$

Q85. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

A normal to the hyperbola, $4x^2 - 9y^2 = 36$ meets the co-ordinate axes x and y at A and B, respectively. If the parallelogram OABP (O being the origin) is formed, then the locus of P is :

(A) $4x^2 + 9y^2 = 121$ (B) $9x^2 + 4y^2 = 169$ (C) $4x^2 - 9y^2 = 121$ (D) $9x^2 - 4y^2 = 169$

Q86. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

A hyperbola passes through the point P$\left(\sqrt{2}, \sqrt{3}\right)$ and has foci at $(\pm 2, 0)$. Then the tangent to this hyperbola at P also passes through the point :

(A) $\left(2\sqrt{2}, 3\sqrt{3}\right)$ (B) $\left(\sqrt{3}, \sqrt{2}\right)$

(C) $\left(-\sqrt{2}, -\sqrt{3}\right)$ (D) $\left(3\sqrt{2}, 2\sqrt{3}\right)$

Q87. AIEEE 2007 · MCQ (Single Correct Answer)

The normal to a curve at $P(x, y)$ meets the x-axis at G. If the distance of G from the origin is twice the abscissa of P, then the curve is a :

(A) circle (B) hyperbola (C) ellipse (D) parabola

Q88. AIEEE 2005 · MCQ (Single Correct Answer)

The locus of a point $P(\alpha, \beta)$ moving under the condition that the line $y = \alpha x + \beta$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is :

(A) an ellipse (B) a circle (C) a parabola (D) a hyperbola

Q89. JEE Main 2023 (Online) 13th April Evening Shift · **Numerical**

The foci of a hyperbola are $(\pm 2, 0)$ and its eccentricity is $\frac{3}{2}$. A tangent, perpendicular to the line $2x + 3y = 6$, is drawn at a point in the first quadrant on the hyperbola. If the intercepts made by the tangent on the x - and y-axes are a and b respectively, then $|6a| + |5b|$ is equal to _____

Q90. JEE Main 2023 (Online) 13th April Morning Shift · **Numerical**

Let m_1 and m_2 be the slopes of the tangents drawn from the point $\mathrm{P}(4, 1)$ to the hyperbola $H : \frac{y^2}{25} - \frac{x^2}{16} = 1$. If Q is the point from which the tangents drawn to H have slopes $|m_1|$ and $|m_2|$ and they make positive intercepts α and β on the x-axis, then $\frac{(PQ)^2}{\alpha\beta}$ is equal to _____.

Q91. JEE Main 2023 (Online) 25th January Morning Shift · **Numerical**

The vertices of a hyperbola H are ($\pm$ 6, 0) and its eccentricity is $\frac{\sqrt{5}}{2}$. Let N be the normal to H at a point in the first quadrant and parallel to the line $\sqrt{2}x + y = 2\sqrt{2}$. If d is the length of the line segment of N between H and the y-axis then d^2 is equal to _____.

Q92. JEE Main 2022 (Online) 28th July Morning Shift · **Numerical**

For the hyperbola $\mathrm{H} : x^2 - y^2 = 1$ and the ellipse $\mathrm{E} : \frac{x^2}{\mathrm{a}^2} + \frac{y^2}{\mathrm{b}^2} = 1$, $\mathrm{a} > \mathrm{b} > 0$, let the

(1) eccentricity of E be reciprocal of the eccentricity of H, and

(2) the line $y = \sqrt{\frac{5}{2}}x + \mathrm{K}$ be a common tangent of E and H.

Then $4\left(\mathrm{a}^2 + \mathrm{b}^2\right)$ is equal to _____.

Q93. JEE Main 2022 (Online) 27th July Evening Shift · **Numerical**

A common tangent T to the curves $\mathrm{C}_1 : \frac{x^2}{4} + \frac{y^2}{9} = 1$ and $C_2 : \frac{x^2}{42} - \frac{y^2}{143} = 1$ does not pass through the fourth quadrant. If T touches C_1 at (x_1, y_1) and C_2 at (x_2, y_2), then $|2x_1 + x_2|$ is equal to _____.

Q94. JEE Main 2022 (Online) 25th July Morning Shift · **Numerical**

Let the equation of two diameters of a circle $x^2 + y^2 - 2x + 2fy + 1 = 0$ be $2px - y = 1$ and $2x + py = 4p$. Then the slope m $\in (0, \infty)$ of the tangent to the hyperbola $3x^2 - y^2 = 3$ passing through the centre of the circle is equal to _____.

Q95. JEE Main 2022 (Online) 26th June Evening Shift · **Numerical**

Let a line L_1 be tangent to the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$ and let L_2 be the line passing through the origin and perpendicular to L_1. If the locus of the point of intersection of L_1 and L_2 is $(x^2 + y^2)^2 = \alpha x^2 + \beta y^2$, then $\alpha + \beta$ is equal to _____.

Q96. JEE Main 2022 (Online) 25th June Evening Shift · **Numerical**

Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $\frac{5}{4}$. If the equation of the normal at the point $\left(\frac{8}{\sqrt{5}}, \frac{12}{5}\right)$ on the hyperbola is $8\sqrt{5}x + \beta y = \lambda$, then $\lambda - \beta$ is equal to _____.

Q97. JEE Main 2021 (Online) 27th August Evening Shift · **Numerical**

Let A (secθ, 2tanθ) and B (secϕ, 2tanϕ), where $\theta + \phi = \pi/2$, be two points on the hyperbola $2x^2 - y^2 = 2$. If (α, β) is the point of the intersection of the normals to the hyperbola at A and B, then $(2\beta)^2$ is equal to _____.

Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **C**	3. **A**	4. **C**	5. **B**
6. **C**	7. **A**	8. **C**	9. **A**	10. **D**
11. **C**	12. **A**	13. **B**	14. **D**	15. **D**
16. **C**	17. **A**	18. **B**	19. **A**	20. **C**
21. **C**	22. **B**	23. **D**	24. **A**	25. **C**
26. **B**	27. **C**	28. **D**	29. **D**	30. **C**
31. **A**	32. **A**	33. **B**	34. **A**	35. **B**
36. **C**	37. **B**	38. **C**	39. **B**	40. **A**
41. **C**	42. **D**	43. **A**	44. **D**	45. **B**
46. **D**	47. **D**	48. **B**	49. **A**	50. **B**
51. **D**				

Numerical

52. **8**	53. **189**	54. **1944**	55. **141**	56. **120**
57. **55**	58. **40**	59. **61**	60. **51**	61. **182**
62. **306**	63. **2**	64. **1552**	65. **88**	66. **42**
67. **2**				

Out of syllabus

68. **B**	69. **D**	70. **B**	71. **C**	72. **D**
73. **C**	74. **D**	75. **B**	76. **D**	77. **D**
78. **B**	79. **A**	80. **D**	81. **A**	82. **D**
83. **C**	84. **B**	85. **D**	86. **A**	87. **B**
88. **D**	89. **12**	90. **8**	91. **216**	92. **3**
93. **20**	94. **2**	95. **12**	96. **85**	97. **Bonus**